

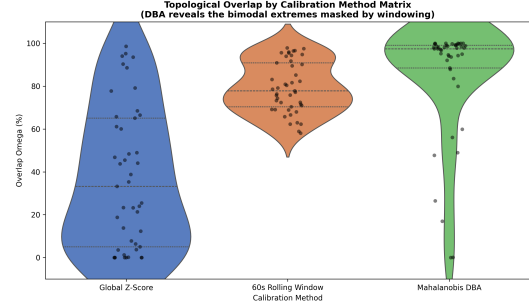
Subject-Specific Covariance-Aware Calibration for Wearable Physiological Sensing Under Drift

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Abstract—Continuous physiological sensing requires robust calibration against hardware-induced signal drift and interpersonal autonomic variance. Standard processing pipelines typically employ generic temporal normalization (e.g., rolling averages) to mitigate these shifts in real-time. In this letter, we evaluate subject-wise static and moving-window normalizations against a Dynamic Baseline Anchoring (DBA) methodology formulated as a multi-dimensional Mahalanobis whitening transformation. Evaluating on 45 subjects across the WESAD and CASE datasets, we implemented blocked cross-validation ensuring exclusive train-split parameter estimation. We empirically demonstrate that continuous rolling normalizations degrade downstream sensor separability (dropping F1 to 0.43 in WESAD). Conversely, we show that subject-specific, covariance-aware calibration prevents the separability collapse introduced by continuous rolling normalization under drift. Our primary contribution is a continuous calibration protocol for wearable sensing. When only a resting anchor is available and rolling correction risks boundary collapse, DBA avoids temporal failure modes while remaining competitive with subject-wise train-block standardization on held-out contiguous blocks.



Index Terms—Wearable Sensors, Sensor Calibration, Data Normalization, Mahalanobis Whitening, Electrodermal Activity.

I. INTRODUCTION

Wearable physiological sensors assert significant potential for continuous objective state monitoring in naturalistic environments [1]. However, deploying these sensors in ambulatory contexts introduces substantial signal volatility, originating simultaneously from dynamic hardware contact (sensor drift) and pronounced intra-subject autonomic variance [2], [3].

To counteract longitudinal drift and standardize inter-subject discrepancies, algorithms typically rely on normalization techniques prior to feature extraction or classification. Standard approaches include subject-wise standardizations [4], [5] and tracking normalizations such as rolling moving-average subtraction [6]. While computationally lightweight, these methods enforce rigid, independent scaling across all signal dimensions, and continuous rolling approaches carry a severe risk of signal erasure: moving averages intrinsically suppress sustained state-changes, potentially destroying the physiological separation required for continuous sensing [5].

Recent analyses indicate that individual deviations in baseline physiology are structurally complex; generic standardizations that indiscriminately compress physiological timeseries into population-derived Gaussian margins may silently distort downstream class separations [6], [7].

In this letter, we present a focused evaluation of *Dynamic Baseline Anchoring* (DBA), structured as a multi-dimensional Mahalanobis-based whitening transformation. Rather than standardizing raw sensor streams to a universal scalar mean, DBA dynamically anchors and decorrelates incoming data blocks relative to the geometric density of the user's individual rest state. By employing strict blocked cross-validation across 45 unique subjects from two public

datasets (WESAD, CASE), we verify that preserving a user's multi-dimensional baseline covariance prevents the temporal F1 collapse commonly caused by moving-average normalizations. Our distinct novelty lies not in proposing a new classification algorithm, but in establishing an empirical wearable calibration protocol: demonstrating that subject-specific baseline-conditioned whitening acts as a leakage-free mechanism to preserve downstream sensor-state separability under continuous drift.

II. METHODOLOGY

A. Active Conditions and Dataset Structuring

We defined the validation task as the continuous, binary separation of a neutral resting baseline against an induced active condition, utilizing two independent datasets partitioned temporally to assess distinct physiological paradigms:

- 1) **WESAD (15 Subjects):** Evaluates a 20-minute resting baseline against an acute stress task via the TSST [4].
- 2) **CASE (30 Subjects):** Continuously tracks individuals viewing affective video protocols [8]. Segments systematically annotated by participants as both high arousal and negative valence were logically grouped to form the active condition against the neutral baseline video segment.

B. Continuous Feature Extraction Space

Raw ECG/BVP and EDA telemetry streams were synchronously resampled to an operational frequency of 100 Hz. We derived dense, continuous physiological states using high-resolution, instantaneous metrics mapping common wearable domains:

TABLE 1. Blocked Test-Set F1 Separability (Mean $\pm$ SD) across Parameter-Locked Normalization Pipelines

Normalization Method	WESAD (Acute Stress)	CASE (High Arousal)
Subject-wise Z-Score	0.90 $\pm$ 0.20	0.74 $\pm$ 0.35
Rolling Z-Score	0.43 $\pm$ 0.23	0.67 $\pm$ 0.15
DBA (Proposed)	0.90 $\pm$ 0.20	0.74 $\pm$ 0.35

- **Continuous RR-derived Wavelets:** Cardiac inter-beat intervals (RR) were tracked continuously. A Continuous Wavelet Transform (CWT) produced instantaneous time-varying spectral coefficients tuned precisely to classic Low Frequency (0.04–0.15 Hz) and High Frequency (0.15–0.40 Hz) structural domains.
- **Instantaneous EDA Components:** Convex optimization (cvxEDA) [9] decoupled the raw stream into tonic (drift) and fast-acting phasic components at 100 Hz.

C. Dynamic Baseline Anchoring via Whitening

Let $\mathbf{x}_t \in \mathbb{R}^n$ represent the raw multivariate sensor column vector at time t . Provided a strictly isolated training baseline block $\mathcal{B}$, we compute the empirical multivariate mean $\mu_{\mathcal{B}}$ and covariance matrix $\Sigma_{\mathcal{B}}$. We formalize DBA mathematically as a Mahalanobis whitening transformation leveraging the fractional power of the inverse covariance matrix:

$$\mathbf{z}_t = \Sigma_{\mathcal{B}}^{-1/2}(\mathbf{x}_t - \mu_{\mathcal{B}}) \quad (1)$$

While equivalent to standard baseline-conditioned whitening, our methodical contribution resides in its execution for drift calibration: estimating parameters exclusively from a training baseline only, and applying it uniformly during contiguous testing to maintain subject-specific covariance without future leakage. This n -dimensional normalized feature space guarantees an equal comparative basis against standardized multivariate vectors.

D. Leakage-Safe Blocked Cross-Validation Protocol

Evaluating high-density (100 Hz) telemetry is extremely susceptible to temporal leakage if samples are split randomly. To guarantee strict independence, we implemented a 50/50 temporal Blocked Split. For each subject, the contiguous first halves of the baseline and active blocks were reserved exclusively for training, while the latter halves were fully held out for linear testing (Logistic Regression, L2 penalty). **All normalization parameters, including the baseline anchors ($\mu_{\mathcal{B}}, \Sigma_{\mathcal{B}}$) and standard Z-score scalars (μ, σ), were estimated on the training portion only and then applied invariantly to the held-out contiguous test portion. For rolling z-score, only past temporal samples were used within the rolling window at any given point.**

III. RESULTS AND DISCUSSION

A. Prevention of Temporal Geometry Collapse

Table 1 indicates that unanchored tracking normalizations (Rolling Z-Score) structurally alter the underlying class separability of extended wearable responses. The 60-second floating window uniformly erased continuous signal offsets, causing the test-set per-subject F1 to drop (0.43 $\pm$ 0.23 in WESAD, 0.67 $\pm$ 0.15 in CASE). Subject-wise

Z-Score provides a strong within-subject reference using combined train-block statistics. However, DBA reaches comparable held-out separability using solely an isolated resting anchor. Subject-wise Z-Score uses combined baseline and active train-block statistics, whereas DBA requires only a resting baseline anchor. This securely mitigates the continuous rolling failure mode without requiring prior exposure to the target active state for scaling variance estimates, yielding robust maintenance across sets (WESAD F1: 0.90 $\pm$ 0.20, CASE F1: 0.74 $\pm$ 0.35). DBA preserved separable boundaries significantly better than rolling normalization in the severe WESAD acute stress task (Wilcoxon signed-rank test, $p < 0.01$).

IV. CONCLUSION

This study evaluated the structural stability of wearable calibration methodologies. We demonstrated that dynamic moving averages suppress sustained offsets at the expense of downstream class separation. Our contribution is a sensor calibration protocol for continuous wearable sensing: by requiring only a single isolated resting baseline anchor, multi-dimensional baseline-conditioned whitening avoids the separability failure modes introduced by rolling drift correction while remaining competitive with train-block static standardization on held-out continuous blocks.

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